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Artificial intelligence for low income countries

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The global adoption rate of artificial intelligence (AI) is rising, indicating its transformative potential. However, this adoption is far from uniform, with low-income countries (LICs) trailing behind significantly. Despite needing AI for development, LICs face multiple challenges in harnessing its benefits, exacerbating existing global disparities in technology adoption. In spite of the potentially important role that AI can play in the development of LICs, AI literature overlooks these countries, with research predominantly focused on more advanced economies. This lack of inclusivity contradicts the principles of distributive justice and global equity, prompting us to explore the importance of AI for LICs, offer a theoretical grounding for AI catch-up, identify effective AI domains, and propose strategies to bridge the AI gap. Drawing insights from the leapfrogging and absorptive capacities literature, our position paper presents the feasibility of AI catch-up in LICs. One crucial finding is that there is no one-size-fits-all approach to achieving AI catch-up. LICs with strong foundations could favor leapfrogging strategies, while those lacking such foundations might find learning and acquisition prescriptions from absorptive capacity literature more relevant. The article also makes policy recommendations that advocate for the swift integration of AI into critical LIC domains such as health, education, energy, and governance. While LICs must address challenges related to digital infrastructure, human capital, institutional robustness, and effective policymaking, among others, we believe that advanced AI economies and relevant international organizations like UNESCO, OECD, USAID, and the World Bank can support LICs in AI catch-up through tech transfer, grants, and assistance. Overall, our work envisions global AI use that effectively bridges development and innovation disparities.

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Introduction

Currently, artificial intelligence (AI) holds a prominent place in discussions. The adoption of AI and its impact on organizations, businesses, and society have been extensively explored (Chui et al., 2018; European Commission, 2017; Abou-Foul et al., 2023; Cooper, 2023; Goodman et al., 2023; Pereira et al., 2023; Polak, 2021; Wang et al., 2023). Remarkably, the IBM Report reveals that the global AI adoption rate reached 35% in 2022, marking a significant four-point surge from the previous year (IBM, 2022). Equally noteworthy is McKinsey's finding that AI adoption more than doubled since 2017 (McKinsey, 2022). Moreover, the emergence of generative AI is poised to amplify this rate, as indicated by multiple sources (McKinsey, 2023; Gartner, 2023; Forbes, 2023).

AI offers numerous benefits and efficiencies to organizations worldwide. These encompass new automation capabilities, improved user-friendliness, and accessibility, as well as a wider array of well-established use cases in both private (Slee, 2020) and public sectors (Khan et al., 2024; van Noordt and Misuraca, 2022). AI finds applications in various domains, ranging from standalone solutions like virtual assistants (e.g., Amazon's Alexa) and large language models (LLMs), such as ChatGPT, with diverse applications (Khan, 2023), to its integration into existing business operations, such as IT applications and processes (Chan et al., 2019; Wan et al., 2022), as well as intelligent finance (Guo and Polak, 2023; Khan and Umer, 2024) and personalized medicine (Schork, 2019), among others. Most AI research also documents that AI has played an important role in achieving several sustainable development goals (SDGs) (Vinuesa et al., 2020).

Despite the significant role of AI in various domains, a major drawback of the predominant research is that AI and its development are centered on the needs, necessities, and values of the high-income countries (HICs), which are at the forefront of AI advancement (Vinuesa et al., 2020). Low-income countries (LICs), defined by the World Bank as those with a GNI per capita of \$1145 or less for 2024–2025 (World Bank, 2024), have been mostly disregarded in AI research and use, despite the evident potential for AI to improve health services, enhance education outcomes, strengthen energy systems, and increase governance efficiency, among other areas. Given the essential role AI can play in LICs, our position article advocates for its adoption by exploring both theoretical and practical perspectives, as well as examining various dimensions of these perspectives and highlighting the significance of LICs in the global AI discourse.

Specifically, we focus on four important aspects of AI in LICs. First, we argue for the importance of AI in LICs, building our argument on the principles of global distributive justice (Cozzens, 2007), technological equity (Manoharan and Carrizales, 2010), and digital decolonization (Jin, 2017). These principles collectively accentuate the imperative of ensuring that technological progress is independently accessible and beneficial to all countries and social strata, based on their unique needs and contexts. Against the backdrop of AI's importance in LICs, we also discuss how AI can potentially mitigate the economic and administrative challenges in LICs, helping them attain wellbeing in a relatively short period. Second, we contend that AI catch-up in LICs is possible by providing theoretical justifications rooted in leapfrogging and absorptive capacities theories. Third, we examine four potential areas in LICs—health, education, energy, and governance—where AI can have a significant impact and be effectively employed to address challenges. We focus on these areas because they constitute low-hanging fruit, with already developed AI tools and systems, and are of urgent necessity in the LIC context, as reflected in the SDGs and the goals of multilateral development institutions. Lastly, we discuss necessary

leapfrogging strategies and absorptive capacities that serve as prerequisites for AI catch-up in LICs.

Our research holds immense practical significance as it sheds light on LICs often excluded from discussions on AI and innovation. We focus on how LICs, frequently confronting development challenges, can bridge the AI gap and enhance their development. Our work demonstrates that although LICs lag in AI adoption, there is hope for them to catch up and harness AI for societal improvement. Imagine LICs gaining access to AI benefits in critical areas like healthcare and education. This potential could help narrow the gap between HICs and LICs, promoting global fairness. The policy objective of our research is to ensure that LICs do not lag in the AI revolution, enabling them to enhance their societies and quality of life.

Our work contributes to the existing literature through several avenues. Primarily, most research on AI has focused on HICs. Diverging from this main discourse, we first make a case for AI in LICs. Despite their dire need for innovation across sectors, these countries lag in AI adoption and thus remain deprived of its potential benefits. Moreover, we examine the importance of AI in LICs, and how it can help such countries attain prosperity, providing a much-needed perspective on AI potential in LICs. Lastly, we discuss four high-impact areas in LICs where the introduction of AI can have a significant positive impact and outline the necessary conditions to realize such an effect. We believe this discourse can help policymakers plan ahead for AI adoption in LICs and motivate researchers to further probe this topic.

The remainder of the article is structured as follows: Section “Background” provides a background on AI inequality between LICs and HICs. Section “Methodology” outlines the methodology of our analysis. Section “Why is AI equally important for LICs” examines the significance of AI for LICs. Section “Is there a theoretical justification for AI catchup in LICs” explores the theoretical rationale for AI adoption in LICs. Section “In which areas can AI be effectively employed within these countries?” delves into potential domains for AI intervention within LICs. Section “How can LICs narrow the AI gap?” outlines policy prescriptions for the adoption of AI in LICs, while section “Conclusion” concludes the paper.

Background

Technology plays an important role in the socio-economic development of countries (Nguyen et al., 2020). Although numerous elements constitute technology, in this paper, we use the term “technology” synonymously with AI, as our paper is set against the backdrop of AI. Given this context, the technological (AI) divide within and across countries can arise due to two main reasons: (1) the lack of software and hardware necessary to develop and use AI and (2) the lack of sufficient knowledge and human training to effectively use AI for societal benefit. Both these aspects of the technological divide draw inspiration from the idea of the ‘digital divide’ proposed by Warschauer in scholarly work on information and communications technology (ICT) (Warschauer, 2004).

In LICs, both AI tools and skilled professionals are scarce compared to HICs (Alonso et al., 2022), contributing to the AI divide between these countries. Existing evidence also shows that current AI applications are focused on HICs, where most AI researchers are located (Vinuesa et al., 2020). Furthermore, to realize AI's transformational potential, over 60 countries—more than 70% of them developed—have published national-level AI policies in the last five years, highlighting the lack of AI-related progress even at the governmental level in LICs (Demaidi, 2023). Moreover, according to the 2022 AI readiness rankings by Oxford

Insights, most governments in LICs are ranked at the bottom, indicating they are the least prepared for AI adoption (Oxford Insights, 2022).

This imbalance extends to academic discourse, as LICs are notably underrepresented in AI research. An OECD report shows that most conference publications on AI emerge from Europe and Central Asia (19%), North America (22%), and East Asia and the Pacific (27%), whereas sub-Saharan Africa accounts for a mere 0.03% in 2020 (Addo, 2023; Zhang et al., 2021). Similarly, the Center for Security and Emerging Technology highlights that AI research publications, citations, and top publishing institutions are dominated by the United States, the European Union, the United Kingdom, and China (Center for Security and Emerging Technology, 2022). Also, researchers from LICs have minor to no role in the international AI discussions held in HICs (Addo, 2023).

Moreover, AI adoption does not adhere to a uniform global pattern; it varies across firms, industries, and geographic regions, with LICs remaining underexplored. For instance, IBM's survey data shows that Chinese and Indian firms exhibit higher AI adoption rates (60%) compared to markets like South Korea (22%), Australia (24%), the United States (25%), and the United Kingdom (26%). While the survey spanned across various countries, it did not consider even a single LIC, suggesting the lack of attention to AI adoption in LICs. Other notable reports also spotlight exceptional prowess in AI research and implementation displayed by only a select few countries. For instance, the United States, the United Kingdom, Germany, and China lead in total AI private investment and the number of newly funded AI companies over the last decade (NetBase Quid, 2022; Stanford University Human-Centered Artificial Intelligence, 2023).

The discussions above point to a clear AI divide between LICs and HICs. Like most other technologies, LICs lag far behind HICs in AI adoption and dissemination. The obvious impact of this divide is going to shape up enhanced differences in labor productivity and incomes between LICs and HICs (Alonso et al., 2022). However, this divide may also influence the growth trajectories of other sectors, such as education, healthcare, energy, and governance. The AI divide is likely to persist in the short and transitional terms and may even extend into the long run (Alonso et al., 2022) if LICs do not proactively devise strategies for smooth and fast AI adoption.

Neglecting LICs in AI discussions runs counter to the principles of distributive justice and technological equity. The principles emphasize a fair and equitable allocation of society's resources, goods, and opportunities, including technologies. The concept of digital decolonization also advocates for the independent use and investment in technologies tailored to the needs and contexts of different countries. Guided by these principles and within the context of global AI discourse, the utilization of AI technology and its benefits should be distributed fairly and uniformly—conditions that are not currently met. Considering this, our research article presents a theoretical and practical case for AI in LICs.

While the future rate of AI adoption in LICs may seem discouraging to some, this paper contends that “AI catching-up” is possible in LICs, as grounded in various economic theories. Drawing inspiration from the leapfrogging and absorptive capacities literature, we put forward policy prescriptions that hold the potential to expedite the swift adoption and integration of AI across several crucial domains within LICs. In essence, our research envisions the potential for uniform and widespread utilization of AI on a global landscape.

Before we make a case for AI in LICs, it is important to discuss the concept of AI itself. The term “artificial intelligence” lacks a universally agreed-upon definition (Ahn and Chen, 2022; Krafft et al., 2020). Nonetheless, researchers have categorized AI into

five prevalent themes in most literature: solving complex problems, emulating human-like processing, exhibiting a degree of intelligence, centering on technological aspects, and managing external data (Hamm and Klesel, 2021). Some also view AI along a continuum, encompassing various technologies ranging from structured, rules-based systems like robotic process automation to unstructured, inference-based systems like machine learning, deep neural networks, and generative AI (Epstein and Hertzmann, 2023; Lacity and Willcocks, 2021). In this article, we take all these broad interpretations of AI without emphasizing any particular type or facet.

Methodology

Our position paper draws on existing literature to advocate for AI in LICs, rooted in traditional economic theory (Abramovitz, 1986) and the technological equity literature (Pathways for Prosperity Commission, 2019; Masters, 2021). We also anchor our approach in the absorptive capacity concept from firm-level management and national-level innovation system literature (Apriliyanti and Alon, 2017; Cohen and Levinthal, 1990; Zahra and George, 2002). Specifically, we benefit from the recent operationalization and deployment of this approach in low and middle-income countries (LMICs) to measure the varying economic responses to similar knowledge inflows in those countries (Khan, 2022a, 2022b, 2022c).

The limited presence of AI in LICs prevents us from conducting a quantitative investigation due to the lack of available data on national AI adoption and usage. Instead, this necessitates a careful but pragmatic analysis of fundamental theoretical considerations for LICs collectively. Although there are AI strategies and domain-specific discussions for certain LICs and developing countries, such as Rwanda, Brazil, and Mexico, with a focus on sectors like healthcare and education, for instance (Demaidi, 2023), we avoid country-level quantitative analysis, as findings may lack validity for other countries and be subject to dynamic evolution.

Even if empirical studies focus on a subset of LICs, the lack of consensus on how to operationalize the AI adoption variable may lead to contradictory results. Additionally, as AI becomes more prevalent, especially with the new wave of generative AI, concerns arise about replicating number-driven empirical methodologies due to its evolving definition and use as it further entrenches in countries. Given that AI is a new and evolving technology, there is limited literature on AI in LICs. To address this gap, we draw on insights from both academic and policy literature, incorporating examples from a broad range of studies.

In addition to the contextual awareness of the authors and team members' advocacy efforts for AI in LICs, we draw on insights provided by anonymous reviewers, editors of the special issue (<https://www.nature.com/collections/ieebagffgg/how-to-submit>), and esteemed faculty members at George Mason University, George Washington University, and Rochester Institute of Technology. We also benefited from feedback provided during the AI and Structural Transformations seminar in June 2024, organized by the Global Evaluation Initiative (GEI), the World Bank, and UNDP. These valuable insights, along with the team's contextual knowledge, established theoretical paradigms, and previous research on absorptive capacity and related themes in developing and poor economies, form the intellectual foundation of our paper.

Why is AI equally important for LICs?

The importance of AI for LICs is underscored by a convergence of pivotal factors. First and central to this significance is the principle of global distributive justice (Cozzens, 2007) and

technological equity (Manoharan and Carrizales, 2010), emphasizing the imperative of ensuring that technological progress, including AI advancement, is available and useful to all countries and sections of the society, irrespective of economic conditions and disparities. These foundational principles position LICs as promising candidates for AI adoption, presenting substantial potential for transformative impacts. This potential aligns seamlessly with the urgent imperative to address SDGs, especially in crucial sectors like health, education, energy, and governance, where AI-driven technological advancements promise marked progress (Adeshina and Aina, 2023; Di Vaio et al., 2020; Holzinger et al., 2021; Tomašev et al., 2020).

In the practical context of LICs, characterized by significant populations that are unserved and underserved, particularly in vulnerable communities in remote areas, there is an immediate need for innovative, scalable, and cost-effective interventions capable of transcending geographical barriers. The potential of AI to provide such smart solutions becomes evident, offering the prospect of delivering essential services to these remote areas.

Furthermore, LICs face economic challenges that require the use of AI-driven digital strategies for economic development. The relationship between technological advancement and economic development has been corroborated by eminent scholars. From Gerschenkron's idea of technological differences (Gerschenkron, 1962) and Kim's concept of "technological capability" (Kim, 1980) to proponents of "new growth theory" (Romer, 1994), technological adoption and technological differences have been central in explaining economic development. Extending this literature to AI advancement, we can assert that embracing AI could catalyze economic growth. This connection is further illuminated by the principle of technological determinism, which posits that technological progress can linearly shape societal developments, including economic outcomes (Dafoe, 2015). Although technological determinism is contested (Cherlet, 2014), in the context of AI's relevance for LICs, the principle would likely suggest that the adoption and integration of AI could deterministically drive economic development and social change, particularly in regions with limited resources and infrastructure.

Moreover, LICs often suffer from corrupt bureaucracies and inefficient institutions (Khan, 2022a), providing low-hanging fruit for the deployment of AI technologies in those countries. AI has the potential to mitigate and counteract the ramifications of corrupt bureaucracies and inefficient institutions in LICs (Köbis et al., 2022; Odilla, 2023). By streamlining processes, enhancing transparency, and optimizing resource allocation, AI can help mitigate the negative impact of institutional challenges as well as strengthen institutions, fostering a conducive environment for growth.

Finally, several other compelling factors underscore the imperative of equitable AI integration in LICs, including economic rent-seeking from AI-dominant countries (Demchak, 2019), the use of Environmental, Social, and Governance (ESG) investing criteria (Pashang and Weber, 2023) as a means of exerting control, and the deliberate exclusion of LIC's AI from Western and Chinese markets (Lee, 2017). These issues are equally relevant in the context of OECD AI Principles (OECD, 2019).

For instance, if loans are tied to meeting ESG standards (Pashang and Weber, 2023), countries may feel compelled to meet these criteria to secure loans, potentially leading to economic dependence and coercion. Relatedly, economic rent-seeking from dominant countries raises concerns about power imbalances shaping the adoption of AI technologies, fostering a new form of dependence between AI-dominant and AI-lagging countries, combining facets of digital imperialism and colonialism (Jin, 2017). This highlights the need for a careful examination of

policies, trade agreements, and global governance structures to ensure that AI development and deployment are equitable and considerate of the diverse needs, goals, and capabilities of all countries. This need is highlighted in several critical discourses on AI (Pathways for Prosperity Commission, 2019; Masters, 2021).

Similarly, the strategic exclusion of LIC AI in Western and Chinese markets raises concerns about potential barriers that LICs may face in accessing and benefiting from such markets for their AI technologies. This exclusion can contribute to a colossal digital (and economic) divide, limiting the equitable participation of LICs in the global AI landscape (Lee, 2017).

In summary, AI's relevance for LICs, guided by distributive justice and technological equity, alongside other factors like economic rent-seeking, the use of ESG criteria for lending, and the exclusion of LIC AI from AI-dominant markets, cannot be overstated. With the potential to drive SDG achievement, serve underserved populations, stimulate economic development, and counteract institutional challenges, AI emerges as a critical technology to usher in a more equitable and prosperous future for these countries.

Is there a theoretical justification for AI catch-up in LICs?

The succinct answer to this question is affirmative. However, before we expound on our response, it would be fitting to touch upon the concept of "catch-up." Commencing with Abramovitz's (Abramovitz, 1986) pioneering work titled "Catching-up, Forging Ahead and Falling Behind," numerous economists, particularly those within the Schumpeterian school, have rigorously examined the phenomenon of catch-up. Scholars such as Freeman (1987), Mathews (1996), Lee (2005), and Mazzoleni and Nelson (2007) have significantly contributed to this discourse. Building on this foundation, Odagiri et al. (2010) assert that catch-up entails the process through which a late-developing country narrows both the income gap ('economic catch-up') and the technological capability gap ('technological catch-up') relative to a leading country. This definition aligns with that of Fagerberg and Godinho (2006).

Expanding upon this body of literature, we define "AI catch-up" as the ability of an AI-impovertised country to diminish the disparity in its AI capabilities vis-à-vis an AI-leading country. Previous literature on technological capabilities underscores their significance in supporting developmental catch-up during the 1980s and 1990s (as evidenced by Mazzoleni and Nelson, 2007; Lee, 2005). In a broader context, we assert that countries fostering AI capabilities and Research and Development (R&D) may narrow their productivity (and, by extension, development) gap with leading countries. Conversely, countries that do not heavily invest in AI capabilities and R&D may fall even further behind.

In the following subsections, we draw on two economic theories, leapfrogging and absorptive capacities, to make a theoretical case for AI catch-up in LICs.

Leapfrogging in AI advancement: a path for some LICs. The concept of leapfrogging, deeply rooted in interdisciplinary fields such as development studies, economics, technology, innovation, and public policy, holds the promise of enabling LICs to bridge the technological divide and achieve AI advancement. Leapfrogging literature offers valuable insights into how disadvantaged countries can strategically leverage their distinct contexts to bypass traditional developmental stages and catapult directly into the forefront of technological progress (Mody and Sherman, 1990; Soete, 1985). This approach allows them to circumvent the challenges and limitations encountered by more developed countries along their historical trajectories.

At its core, leapfrogging entails the ability of less developed or disadvantaged countries to leap over conventional stages of development and directly adopt advanced technologies or practices designed by developed countries. This phenomenon empowers them to catch up with or even surpass their more developed counterparts. The increasing tendency toward globalization makes leapfrogging even more plausible, as analyzed by studies in the case of digital TV (Lee et al., 2005).

Historically, cases of technological leapfrogging have been well-documented. Scholars such as Alexander Gerschenkron have emphasized the potential for latecomer countries to attain rapid economic growth by harnessing advanced technologies (Gerschenkron, 1962). These technologies enable them to bypass the constraints (Lee, 2005), as faced by early industrializers. Given that many early entrants have established operations within the industry by the time latecomers arrive, latecomers can glean insights from both the triumphs and failures of these pioneers, allowing them to better identify the sources of competitive advantage within the industry (Csaszar and Siggelkow, 2010). Other studies shed light on the pivotal role of ‘innovation systems’ and technological learning in facilitating successful leapfrogging (Kim, 1997; Nelson, 1993). Yet studies on diffusion theory elucidate how innovations can quickly propagate within societies, leading technologically lagging countries to embrace the most current technologies (Rogers, 2010).

Extending this discourse to AI, leapfrogging AI advancement for LICs emerges as a formidable but feasible pursuit. AI technologies can catalyze this leap with their potential to address pressing challenges unique to LICs. Historical instances of technological leapfrogging, where countries leapfrogged developmental stages by adopting new technologies, provide a basis for considering the possibility within the realm of AI. Evidence shows that technologies like mobile-based e-commerce and e-banking have been adopted faster in low- and middle-income countries (LMICs) compared to HICs (Novartis, 2020), supporting the idea that some LICs can leapfrog in AI adoption with the right conditions.

However, it is crucial to acknowledge the challenges inherent in this endeavor. The research underscores that while technological catch-up is attainable, it necessitates meticulous planning, investments in human capital, and policy interventions. The absence of requisite digital infrastructure, skilled workforce, and research capabilities often hinders direct AI advancement pathways for LICs. Moreover, the “digital divide,” encompassing challenges like brain drain, limited access to data, and insufficient research funding, can further compound the technological disparity between AI leaders and low-income countries. Due to these disparities, we contend that leapfrogging may not be the most suitable strategy for all LICs; most LICs lack the necessary foundational capabilities to imitate and iterate AI leaders’ current technologies, business models, and product offerings. Hence, leapfrogging may not prove a strong strategy for those countries. However, countries with a strong capability base may opt for leapfrogging. Other LICs may focus on developing their capability base and acquiring learning from the experience of AI leaders, as pleaded by absorptive capacity and other path-following literature.

Nonetheless, noteworthy success stories such as Estonia’s digital transformation and Rwanda’s AI-driven health initiatives (Malhotra, 2023) underscore the potential of leapfrogging. These examples highlight the effectiveness of visionary leadership, strategic investments, and context-specific solutions. Collaborative partnerships with international organizations and AI pioneers also play a pivotal role in bridging knowledge gaps and facilitating knowledge transfer.

Absorptive capacity: a path for most LICs. In contrast to direct imitation, absorptive capacity emphasizes learning from the experiences of others, aligning with path-following literature (Lee and Lim, 2001). Instead of directly imitating the current state of leading countries, LICs follow the sequence of steps taken by technological leaders. In other words, absorptive capacity literature views LICs as dynamic learning entities (Khan, 2022c). Broadly, it pertains to a country’s ability to effectively acquire, assimilate, and apply new knowledge and technologies (Cohen and Levinthal, 1990; Zahra and George, 2002). The approach necessitates enhancing local capacities such as institutions, policies, and infrastructure to attract technological inflows from abroad (Khan, 2022a). Essentially, this approach requires LICs to strategically advance their technological capabilities by integrating themselves better with advanced economies (Chen and Sun, 2023).

When applied to the domain of AI, LICs would traverse the developmental trajectory followed by AI leaders. This endeavor would encompass studying the evolution of AI, scrutinizing past experiences, identifying the stages of AI product and process development, and assessing strategies for market entry, among other essential steps. Such an approach would furnish a structural roadmap for the learning journey of LICs, mirroring the path previously paved by AI leaders. Through more efficient integration with advanced economies, LICs can accumulate AI capabilities. This approach can effectively guide LICs in adopting AI technologies developed elsewhere, emphasizing the importance of drawing insights from external sources and adapting these technologies to suit local contexts.

By leveraging insights from absorptive capacity literature, most LICs can systematically approach the development of AI capabilities, focusing on learning, adaptation, collaboration, and innovation. This approach positions them to surmount challenges and harness the benefits of AI for sustainable development.

While the absorptive capacity approach for AI in LICs holds promise, it comes with challenges. Effective knowledge assimilation and learning require strong institutions, skilled human capital, and tailored policies. Adapting technologies to local contexts faces hurdles due to varying socioeconomic factors. Collaborative partnerships with advanced economies can raise dependency concerns. Lastly, sustaining absorptive capabilities demands ongoing investment, which could strain limited resources. Addressing these challenges is vital to harnessing the approach’s potential for fostering AI advancement in LICs.

Is the path-following approach, as suggested by absorptive capacity, applicable to all LICs? Many LICs lack the foundational capacities needed to advance their AI local stocks. Therefore, all these LICs should strengthen their local capacities to attract external expertise in AI. As most LICs struggle with their requisite capacities, absorptive capacity approaches are particularly relevant to their circumstances.

In which areas can AI be effectively employed within these countries?

AI can be impactful in various domains; several studies show that developing countries emphasize leveraging AI in education, healthcare, and government (Ahmed et al., 2022; Demaidi, 2023; Guo and Li, 2018). This study has chosen to focus on the health, education, energy, and governance sectors due to their pivotal roles in the progress and sustainable development of LICs, which often encounter significant challenges within these domains. These domains are also integral components of the SDGs (<https://www.un.org/sustainabledevelopment/blog/2015/12/sustainable-development-goals-kick-off-with-start-of-new-year/>), making them priority areas for multilateral institutions.

Moreover, the tools developed by high-income countries, specifically for healthcare and education, can be easily and promptly transferred to LICs with minor context-based changes, primarily because the nature of problems in these domains is almost similar in both HICs and LICs. For instance, in the health domain, more than 2 billion chest X-rays are performed worldwide every year, and in some countries, AI is being used to detect pneumonia through chest X-rays (Topol, 2019). Interestingly, AI has been found to outperform human radiologists in several instances. As X-rays are a standard medical procedure performed in all countries, AI tools developed to process X-rays and diagnose diseases can be easily transferred to LICs, potentially creating a significant impact on disease detection and the speed of processing X-rays. Similarly, AI is employed in higher education to provide personalized learning paths to students, support collaborative learning via intelligent systems, and offer virtual reality experiences (Zawacki-Richter et al., 2019; Capraro et al., 2024). All these tools are equally relevant to students in LICs, and perhaps can be transferred without huge alterations. Similarly, we expect the AI tools in governance and energy developed for HICs can be easily ported to LICs. This is another reason we focused on these four domains.

While we acknowledge that there are multiple domains where AI can have a profound impact in LICs, the identification of all these domains is not the primary theme of our work. Therefore, we focus here on some of the promising and important domains.

Health. Health outcomes in LICs are suboptimal, and the COVID-19 pandemic has further exacerbated the challenges faced by their already struggling healthcare systems (Hamid et al., 2020; Kaye et al., 2021). One of the evident challenges in the healthcare domain is an unequal distribution of healthcare services and resources in rural and urban areas. Major city hospitals like those in Addis Ababa, Mogadishu, Kabul, and even Islamabad are overwhelmed, while rural medical facilities lack resources. Some of these rural hospitals are inactive, lacking healthcare professionals especially female medical staff. On the socio-cultural front as well, healthcare is affected by societal norms. For example, in certain countries, such as Afghanistan and Pakistan, few female doctors continue to practice after marriage (Masood, 2019). These societal norms contribute to a gender disparity in healthcare administration, as many female patients prefer consultations with female practitioners, especially in conservative rural areas.

The present advancement in AI offers a window of opportunity to develop solutions to the aforementioned healthcare challenges in LICs. For instance, AI technologies can be leveraged to promote telemedicine in some LICs, which can potentially close the gendered and geographic gap in public health. Telemedicine can extend medical services to remote areas by utilizing untapped female doctors. In LICs with higher smartphone adoption rates, AI-powered telemedicine can naturally address the growing demand for medical care.

In other regions, AI can enhance healthcare supply chains. For example, autonomous drones can distribute medicine to remote African areas. Countries like Rwanda, Ghana, Nigeria, and the Ivory Coast have partnered with California-based Zipline to lead in this effort (Appio et al., 2023; Damoah et al., 2021). Zipline delivers 75% of Rwanda's blood supply outside of the capital (Nisingizwe et al., 2022; Reed, 2022). Other countries such as Malawi, Senegal, and Madagascar have also utilized autonomous drones to strengthen healthcare provision (Knoblauch et al., 2019). These drones have reduced maternal mortality and malnutrition by ensuring timely medical supplies and minimized carbon emissions by saving hospital commute time.

Additional applications of AI in LICs encompass clinical decision support systems (Wang et al., 2023), treatment planning (Ugarte-Gil et al., 2020), triage assistance, and health chatbots (Limited Index Labs. T. C., 2018; Montenegro et al., 2019). Compared to HICs, the utilization of AI in LIC healthcare is in its infancy (Wahl et al., 2018). Deploying AI tools in LICs necessitates overcoming challenges like limited data availability, building trust, and demonstrating cost-effectiveness (Ciecierski-Holmes et al., 2022). Despite current challenges, AI-driven emerging technologies hold tremendous promise for transforming the provision of healthcare services in LICs. Overcoming these challenges could lead to even further improved health outcomes in LICs' settings.

Education. Education outcomes in LICs are markedly worse. UNESCO reports that approximately 258 million children, adolescents, and youth were out of school in 2018, equating to about one-sixth of the global school-age population aged 6–17 (UNESCO, 2019). The pandemic has further exacerbated this situation, profoundly impacting learning even in more affluent economies, not to mention LICs (Engzell et al., 2021). LICs, already challenged by inadequate resources and infrastructure, have experienced aggravated education setbacks (Betthäuser et al., 2023). According to a joint estimate from the World Bank, UNICEF, USAID, and other partners, learning poverty has increased by a third in LMICs (World Bank, 2022).

Education provision in LICs also suffers from gender and geographic disparities. Many female children remain excluded from schooling in Sub-Saharan Africa (Browne and Barrett, 1991) and other low-income countries (Latif, 2009). Early marriage among female students truncates their educational pursuits (Arthur et al., 2018), intensifying the issue. Furthermore, remote areas in many LICs have “ghost” schools with inadequate infrastructure and absent teachers (Chaudhury et al., 2006; Gilani, 2013).

AI has the potential to alleviate these educational challenges in LICs. For instance, AI-assisted technologies can revolutionize the educational landscape within resource-constrained settings. Diverse instructional formats such as distance learning, personalized instruction, and online learning can be made accessible in LICs through AI applications. Moreover, AI can be a critical tool in supporting educational systems in LICs by aiding teachers in devising effective study plans and selecting appropriate learning methodologies. The success of AI's integration and other complementary technologies into education is already evident in the experiences of several LICs. In many LICs, for instance, RoboTutor is an AI-powered educational platform that offers personalized tutoring in reading and math to children, aiming to address the shortage of teachers in rural areas (Goswami et al., 2019). Similarly, mobile Edtech solutions are utilized in Sub-Saharan Africa (Oberdieck, 2021). In Africa, the African Virtual University utilizes AI-driven chatbots to support students in their studies and help bridge the educational gap (African Development Bank, 2019).

However, there are challenges in implementing AI in education within LICs. Limited access to technology and the internet remains a significant barrier. In many LICs, basic infrastructure like electricity is lacking, making widespread AI technological adoption difficult. Moreover, building trust in AI-based systems and overcoming cultural resistance to change are hurdles that need to be addressed (Holmes and Porayska-Pomsta, 2022). Additionally, the scarcity of high-quality educational content in local languages poses a challenge for effective AI integration. Financial constraints and the need for proper teacher training further complicate the adoption of AI solutions in LICs.

Given these formidable challenges, we propose a strategy that involves partnering initiators with individuals or organizations from representative LICs. These collaborators will be equipped with foundational knowledge about AI, supplemented by familiarity with pre-existing AI-based systems or applications. Simultaneously, policy promotion is advocated, emphasizing partnerships between LICs and enthusiastic stakeholders. By capitalizing on platforms like Facebook and YouTube, commonly utilized across countries, the approach initiates by acclimatizing participants to a familiar AI-based apps or systems. This introductory phase illuminates the potential of AI in enhancing education and shaping careers. A critical dimension of this strategy is to empower girls and women with AI knowledge and skills, thereby contributing to the amelioration of societal inequalities in the tech realm.

In conclusion, while AI holds promise to transform education in LICs, complex challenges need to be navigated for its successful implementation. AI can play a pivotal role in improving education outcomes in these settings by addressing infrastructure issues, cultural adaptation, content localization, and teacher training.

Energy. LICs often face energy challenges linked to energy production, storage, and distribution, as well as an imbalance between the supply and demand for energy. Antiquated infrastructure, particularly aging grid systems, causes distribution losses, chronic blackouts, and other production and distribution inefficiencies. Reliance on fossil fuels, alongside intermittent hydropower sources, also complicates the supply-demand imbalance (Palensky and Dietrich, 2011). Long-standing flawed energy policies, bureaucratic inefficiencies, and poor future planning further compound these challenges. Moreover, heavy reliance on oil-based energy generation sources also degrades the environment and puts LICs in a more disadvantageous position. At the core of these issues are primarily faulty planning and irrational choices exercised by individuals in charge of energy production and management systems.

'Smart Energy' can be an excellent AI-based solution to the abovementioned problems. Smart energy, characterized by smart grids, combines AI, big data, cloud computing, and the Internet of Things (IoT) to optimize energy production and its distribution to households and commercial facilities, minimize grid losses, eliminate the supply-demand gap via precise planning and projections, hence ensuring affordable and reliable energy (Cheng and Yu, 2019). Kenya offers a notable example: companies like SteamaCo utilizing AI-powered smart meters and distributed grid systems to ensure reliable and affordable electricity delivery to remote regions (Ashden Climate Solutions in Action, 2015).

At the macro level, AI algorithms can improve energy efficiency in LICs by analyzing real-time data from energy systems, thus optimizing consumption patterns and distribution. AI algorithms can also predict national renewable energy production from sources like solar power, aiding utilities in planning their seamless integration into the grid. Moreover, predictive maintenance powered by AI can also identify potential failures in power plants or transmission lines, leading to enhanced longevity of energy infrastructure and reduced downtime. At the micro level, AI-based smart energy solutions can transform energy consumption in LICs. For example, using smart meters, consumers can be frequently updated about their energy consumption patterns and associated costs, nudging them to make better energy consumption choices (Saidani Neffati et al., 2021). Similarly, AI-managed cooling, heating, and illumination

systems in buildings can reduce energy consumption and lower emissions (Mehmood et al., 2019).

While AI-based tools have a substantial potential to mitigate the energy problems of LICs, it is important to note that these tools require essential infrastructure. For example, smart energy requires robust digital infrastructure, including smart grids and smart meters. Furthermore, the public must be trained to understand the importance and operability of smart energy tools (Saidani Neffati et al., 2021). All these infrastructural and educational interventions would require significant investments from LIC governments. Despite these obstacles, the implementation of such solutions remains achievable. By adopting a strategic approach that prioritizes international collaboration, focused investments, and innovative thinking, LICs can effectively leverage both AI and their abundant natural resources to surmount energy deficiencies and drive sustainable development. This transformative process has the potential to enable LICs to bypass traditional energy infrastructure and develop efficient, clean, and accessible energy systems that cater to the needs of their populations.

Government administration. Citizens in every country rely on their government for various benefits and services. As such, citizens' well-being and trust in the government are closely tied to the type and quality of services provided by local or state authorities (Mehr, 2017). However, governments, particularly in LICs, often face multifaceted challenges. These challenges include inefficient and slow bureaucratic processes, pervasive corruption, and a lack of proper accountability. Additionally, many governments in LICs continue to rely on manual systems that involve extensive paperwork, which poses several issues. Firstly, the heavy reliance on paperwork burdens the workforce, potentially affecting the speed and quality of their work (Mehr, 2017). Secondly, paper-based systems hinge upon essential data collection, which could otherwise be employed to develop efficient governance tools and policies (Pencheva et al., 2020).

Most of the challenges mentioned above are embedded into government systems that heavily rely on human-to-human interaction and paper-based documentation for repetitive tasks. AI-based automation of most such government processes can enhance efficiency, enabling the delivery of services to a greater number of citizens in a shorter time. Moreover, this automation can also enhance citizen engagement and trust, promote interoperability and accounting, and free the government's labor for other tasks where human-to-human interaction is inevitable (Sharma et al., 2020; Zuiderwijk et al., 2021). The existing evidence from low-income (and low-middle-income) countries supports this argument. For example, the Indian state of Karnataka implemented an electronic system of land records that eliminated 1.32 million days of wait time for seven million farmers (Adam and Fazekas, 2021). Similarly, the Egyptian government introduced e-health services to conduct breast cancer tests for women nationwide, helping diagnose and treat breast cancer. The Nigerian government introduced an e-agriculture system, providing strategic information to the farmers and promoting IT skills (Stoiciu, 2011).

Financial corruption within government institutions is another major problem that hinders progress, increases citizen dissatisfaction, and obstructs the timely delivery of essential services. While corruption is not confined to LICs, most countries on the list of the worst corruption perceptions are LICs (Transparency International, 2022). AI offers immense potential to mitigate corruption in government institutions. Often, substituting government officials who frequently engage with citizens with automated and tightly controlled e-governance tools can mitigate

corruption and enhance government revenues. The evidence from LICs (and low-middle income) lends credence to this proposition. For example, the introduction of an electronic land record system in the Indian State of Karnataka eliminated the bribe that farmers had to pay to government officials for obtaining records, resulting in saving 806 million Indian Rupees in potential bribes (Adam and Fazekas, 2021). Similarly, the introduction of mobile payment for urban water bills reduced corruption and increased government revenues in Tanzania (Krolikowski, 2014).

Another potential and impactful area where AI can transform governance in LICs is data-driven analytics and policymaking. As transactions between citizens and governments generate massive amounts of data, it can be used to offer personalized services. For instance, AI tools, utilizing existing data, can formulate educational programs for school-going children that have yielded positive outcomes for similar demographics. AI can also be used to accurately forecast and simulate specific events by using machine learning algorithms. For example, AI can anticipate future medical and educational requirements, the likelihood of floods and traffic accidents, and the need for emergency services (Margetts and Dorobantu, 2019). These projections and forecasts are even more critical to LIC governments, who generally are resource-deficient. By using AI-driven initiatives, governments in LICs can increase the accuracy of their interventions and save the unnecessary waste of resources that faulty human projections can cause.

We briefly discussed the potential and high-impact avenues of AI-based governance in LICs. However, this list is by no means exhaustive. Overall, the judicious application of AI in governance can have a substantial positive impact on the efficiency of the government, can mitigate corruption, and enable governments to forecast and simulate events with high precision.

How can LICs narrow the AI gap?

In the preceding sections, we have presented the theoretical rationale for LICs to narrow the AI gap and bolster their AI capabilities. Our discussion has been rooted in both the leapfrogging and absorptive capacity frameworks. Moreover, we have underscored pivotal domains where AI applications can bring about transformative changes within LICs. In this section, we put forth a series of actionable strategies to guide LICs in their pursuit of bridging the AI divide. These strategies are a direct outgrowth of the arguments elucidated in the section “Is there a theoretical justification for AI catch-up in LICs?”. Specifically, these strategies constitute a multifaceted approach that draws inspiration from both the leapfrogging and absorptive capacity literature.

Embracing leapfrogging strategies. For some LICs, the leapfrogging approach offers a compelling policy-driven strategy. Rooted in innovation and agile adaptation, this strategy envisions LICs bypassing intermediate stages of technological development to integrate advanced AI solutions directly. In alignment with this approach, LICs can strategically identify entry points within AI domains that promise immediate and substantial developmental impact. Under leapfrogging strategies, we suggest a few below with examples from LICs (and LMICs) who followed these strategies.

Creation of agile technological infrastructures. An agile technological infrastructure involves developing systems that can quickly adapt to technological changes. This includes investing in the availability of high-speed internet, including 5G infrastructure, robust data storage, and processing capabilities, such as data management systems capable of handling large volumes of

data for AI training, and cloud computing for scalable and cost-effective infrastructure for AI applications like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform. The objective of such infrastructures is to create an environment where LICs can seamlessly incorporate and upgrade cutting-edge AI solutions and applications. An example of inspiration is the Rwandan government, which has dedicated efforts to improve internet connectivity and establish tech hubs (Obeysekare et al., 2017). The Kigali Innovation City project, in particular, aims to provide a conducive environment for technology-based innovation, including AI research and development (Burns, 2021).

Fostering context-driven innovation. Fostering context-driven innovation is another pertinent aspect of the recipe for AI adoption. It involves tailoring solutions to address and understand unique challenges, needs, and opportunities within a country. This process requires understanding the local context, including cultural, economic, and social factors, to develop AI applications that are relevant and effective. One such example of context-driven AI innovation is Brazil, which has focused on applying AI in agriculture, considering its importance to the economy (Pivoto et al., 2018). Embrapa, the Brazilian Agricultural Research Corporation, has used AI to optimize agricultural practices, taking into account the diverse ecosystems and challenges faced by Brazilian farmers.

Leveraging digital platforms and open-source AI libraries. This strategy involves utilizing existing digital platforms and open-source AI libraries to access AI knowledge and tools effectively. Open-source libraries provide a foundation for building AI applications, reducing development time and costs. For instance, Kenya has successfully embraced digital platforms for financial services, exemplified by M-Pesa (Kingiri and Fu, 2020; Ngugi et al., 2010). Leveraging these platforms has not only transformed the financial landscape but also created a technological foundation for broader digital innovation, including potential AI applications.

Expedited learning curve through online resources. Capitalizing on online resources involves promoting online education, training programs, and communities to enhance the skills of the workforce in AI-related fields. This strategy accelerates the learning curve, allowing individuals and organizations to quickly acquire the knowledge needed for AI development and implementation. For example, Nigeria has seen a surge in online education platforms and tech communities (Eze et al., 2018). Organizations and educational institutions have encouraged the use of online courses and resources to build a skilled workforce in areas such as software development and AI.

Piloting initiatives to showcase practicality of AI. Piloting initiatives involve testing and demonstrating the practical applications of AI in specific sectors. These initiatives serve as a proof of concept, building confidence among stakeholders and encouraging broader integration efforts. An illustrative example comes from Latin American countries, which have initiated pilot projects to showcase the effectiveness of AI in addressing societal challenges, particularly in healthcare (Chatterjee et al., 2020; García Alonso et al., 2022). By applying AI to improve patient outcomes, these initiatives pave the way for the broader adoption of AI solutions in the country.

These strategies collectively empower LICs to catch up with technological advancements and leapfrog certain stages of development. By combining infrastructure development, context-driven innovation, digital platform utilization, expedited learning through online resources, and practical pilot initiatives,

these countries can position themselves as active participants in the global AI landscape. Each strategy plays a crucial role in creating an ecosystem that fosters innovation, adaptation, and sustainable AI growth.

Harnessing absorptive capacity. In contrast, many LICs can benefit from policy-driven prescriptions that align with the absorptive capacity framework. This approach entails a nuanced process of acquiring, assimilating, and applying new knowledge and technologies acquired from outside sources, adapted to the local context. Of many prescriptions under this approach, we suggest some below.

Strengthening institutional frameworks. This element underscores the significance of robust institutional frameworks. These frameworks create an ecosystem that fosters AI advancements by providing support for research, development, and integration. Investing in institutions dedicated to AI exploration is fundamental for long-term success. An example is that of Singapore's commitment to building a robust institutional framework, such as the AI Singapore initiative, which showcases the importance of dedicated institutions in fostering AI capabilities (Teddy-Ang and Toh, 2020).

Policy-driven investments in human capital. Furthermore, the absorptive capacity approach calls for policy-driven investments in human capital through educational programs and partnerships with academic institutions. By cultivating a skilled workforce, LICs can fortify their ability to drive AI advancements. For instance, LICs can take inspiration from Estonia, which focuses on educational programs, including partnerships with universities, demonstrating the impact of human capital investment in developing a workforce adept in AI technologies (Pihlak, 2019).

Tailored policies for unique contexts. The absorptive capacity approach also advocates for tailored policies designed to suit the unique socio-economic contexts of LICs. These policies are essential to encourage AI-focused research, innovation, and collaboration. By aligning AI development strategies with local challenges, these policies amplify their impact and foster sustainable progress. For example, India's policies that address local challenges, such as the National Mission on Interdisciplinary Cyber-Physical Systems and smart cities, illustrate the effectiveness of tailoring strategies to the country's unique context (Ahmad et al., 2021).

Collaborative Networks with Advanced Economies. Absorptive capacity also mandates the establishment of collaborative networks with advanced economies and international organizations. These networks facilitate knowledge exchange and technology transfer, thus accelerating the learning curve for LICs (Wagner et al., 2001). LICs may follow South Korea's path in this regard. South Korea's collaborative efforts with advanced economies, exemplified by partnerships with research institutions and industry leaders, showcase the benefits of international networks in advancing AI capabilities (Herrador, 2023).

Facilitating technology transfer. Additionally, the absorptive capacity approach emphasizes the importance of policies that facilitate technology transfer. Such policies are pivotal in enabling LICs to adopt AI technologies from HICs and tailor them to local contexts. By strategically aligning themselves with global collaborators, LICs can harness absorptive capacity as a policy-driven mechanism to enhance their AI capabilities. In this regard, Mexico's initiatives that focus on technology transfer, including

partnerships with global tech companies, highlight the effectiveness of policies in adopting and adapting AI technologies (Mexico Business News, 2019).

In sum, the absorptive capacity approach provides LICs with a comprehensive policy toolkit that encompasses institutional strengthening, human capital investment, tailored strategies, collaborative partnerships, and technology adaptation. This multifaceted approach empowers LICs to navigate the intricate AI landscape with strategic intent and precision, ultimately closing the AI gap.

Conclusion

The burgeoning global enthusiasm for AI adoption, spanning from initiatives such as the Pan-Canadian AI Strategy, the EU AI Strategy, and the UK AI Safety Institute to the American AI Initiative and China's National Strategy for AI, indicates a transformative era. While countries worldwide recognize AI's transformative potential (Fatima et al., 2020), the journey toward AI integration remains uneven, as LICs find themselves trailing behind. In the entire African continent, only a few countries, such as Rwanda, Ghana, and Tunisia, have developed national AI strategies (World Bank, 2021). Despite their urgent need for innovation to achieve the SDGs, LICs encounter significant barriers that impede their ability to access the potential benefits of AI. These barriers include insufficient digital infrastructure, a lack of accurate digitized data, a limited AI talent pool, and a lack of institutions and regulations governing data privacy and safety, among other challenges. The resulting low AI capabilities in LICs further deepen existing global inequalities.

Additionally, the prevailing AI discourse in literature exhibits a stark imbalance, with a noticeable need for more attention to LICs. Similarly, the control of AI markets and critical resources, such as chips for AI systems, by a select few countries makes LICs less autonomous in their AI investments. Such gaps and disparities raise questions about equity and justice, motivating us to delve into the multifaceted dimensions of AI and its significance for LICs.

Our analysis provides a theoretical foundation for LICs to bridge the AI gap, delineates effective domains for AI implementation, and proposes strategies that could enable these countries to overcome their AI-related adoption challenges. At the heart of this investigation is a resounding assertion: the notion of "AI catch-up" is not a distant dream for LICs. Grounded in the established paradigms of leapfrogging and absorptive capacities literature, this assertion signifies a beacon of possibility. Moreover, we present policy prescriptions that advocate for AI's rapid and comprehensive integration into pivotal domains such as health, education, energy, and governance within LICs. In essence, this work envisions an inclusive global landscape, where AI becomes a unifying force that bridges development disparities, fosters innovation, and drives sustainable progress.

While LICs may appear homogenous from the outside, they are far from being a monolithic entity. In reality, they exhibit significant disparities in institutions, policies, and available resources. Therefore, it is reasonable to assert that there is no one-size-fits-all approach to achieving AI catch-up. For instance, a LIC endowed with robust foundational capacities might be well-suited for leapfrogging strategies, while one lacking such capacities may find the recommendations from absorptive capacity literature more relevant.

It is also crucial to emphasize that AI catching up is a gradual process, not an immediate achievement. To succeed in this endeavor, LICs need to address challenges related to digital infrastructure, human capital, institutional robustness, and effective policymaking. Notably, LICs must establish platforms to

foster AI research, development, and commercialization, all of which will contribute to the growth of their AI capabilities. Furthermore, they must remain adaptable and agile in their AI development strategies, constantly refining them based on experiences and outcomes. Finally, LICs should invest in AI-related research, promote innovation hubs, and support local AI startups to strengthen their AI ecosystem. This concerted effort will help cultivate their domestic AI capacity and shape development strategies tailored to their specific challenges, needs, and industries.

Moreover, we should note that AI catch-up, as suggested in this work, will be executed in collaboration with advanced AI economies. Given that LICs are so far from the technological frontier, AI catch-up in isolation would remain a distant aspiration. Therefore, advanced economies in AI and other international organizations such as UNESCO, OECD, USAID, and the World Bank must play a pivotal role in supporting LICs to achieve AI catch-up through various means, including technology transfer, technological cooperation grants, and technical assistance mechanisms. Creating an environment that fosters knowledge transfer, research collaboration, access to AI-related resources, and personnel training will be more critical than ever. LICs must also actively seek collaboration and engagement with international organizations, academia, and tech companies to access AI expertise, resources, and knowledge-sharing platforms.

As with any research endeavor, our work remains inconclusive. Given that AI diffusion and adoption are not deeply rooted in LICs, our study primarily offers an ex-ante investigation. Over time, an ex-post analysis of AI adoption processes in LICs could either validate or contradict the findings presented here. Moreover, while our study is firmly anchored in established theories, it requires a thorough empirical examination. If future empirical analysis fails to support the proposed AI catch-up, researchers can shift focus to identify the hindering factors and propose interventions. Notwithstanding these constraints, the primary objective of our work is to amplify the voices of LICs within the global AI discourse, and it should be interpreted as such.

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MSK: Conceptualization, methodology, supervision, writing—original draft preparation, writing—reviewing and editing. HU: Investigation, writing—original draft preparation, writing—reviewing and editing. FF: Validation, writing—reviewing and editing.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was not required as the study did not involve human participants.

Informed consent

This article does not contain any studies with human participants performed by any of the authors. Consequently, the need for obtaining informed consent from human participants was not applicable in the context of this research.

Additional information

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